NC STATE UNIVERSITY
Electrical and Computer Engineering

Robust Federated Q-Learning with Almost No Communication

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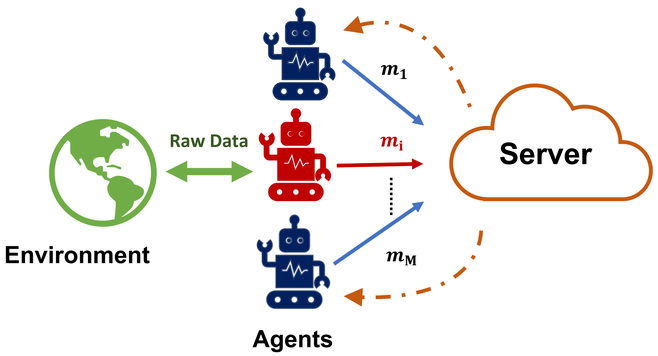
Model and Problem Formulation



Consider an MDP $\mathcal{M} = (S, A, P, R, \gamma)$.
We want to **learn a policy** μ that maximizes

$$V_\mu(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | s_0 = s, \mu\right]$$

❑ Q-Learning [1] solves this problem with finite-time rate $\tilde{\mathcal{O}}\left(\frac{1}{\sqrt{T}}\right)$.
How to improve sample efficiency ? Multi-Agent Collaboration.

Number of Agents: **N** ⬆ **Faster Learning** $\tilde{\mathcal{O}}\left(\frac{1}{\sqrt{NT}}\right)$ ⬆

Problem Setting:
- N agents; a fraction $\boldsymbol{\varepsilon}$ of the agents are adversarial.
- Adversarial agents are omniscient and can act arbitrarily.

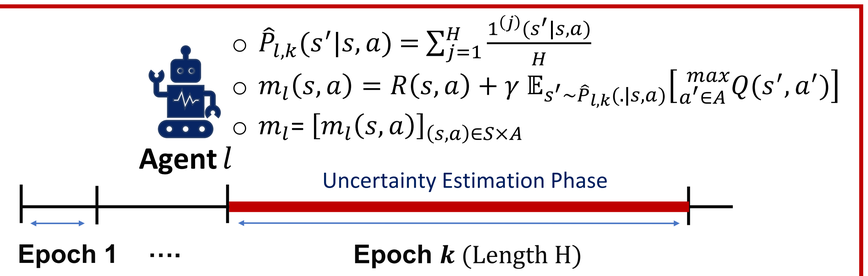
Goal:
- To reliably estimate the optimal policy μ in the federated multi-agent RL setting by pooling experience across many agents, even when a fraction of agents can corrupt the learning signals.

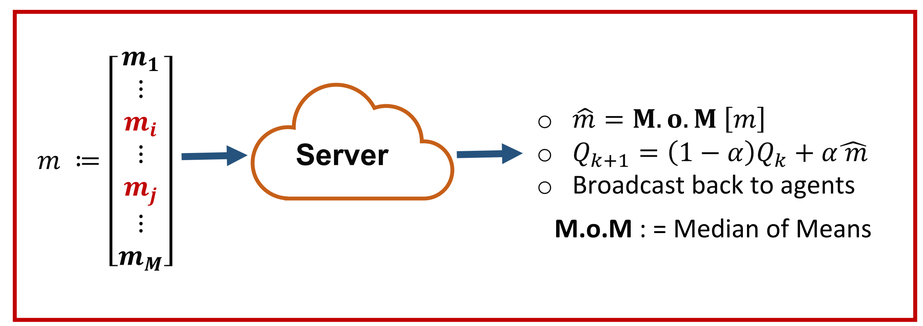
Contributions:

❑ Propose a robust algorithm called Robust Fed-Q [2].
❑ Guarantees finite-time rate of $\tilde{\mathcal{O}}\left(\frac{1}{\sqrt{NT}} + \sqrt{\frac{\varepsilon}{T}}\right)$.
❑ Achieves **collaboration benefit**, and **vanishing corruption** effect.

Robust Fed-Q

Challenge: Account for statistical uncertainty + adversarial behavior.





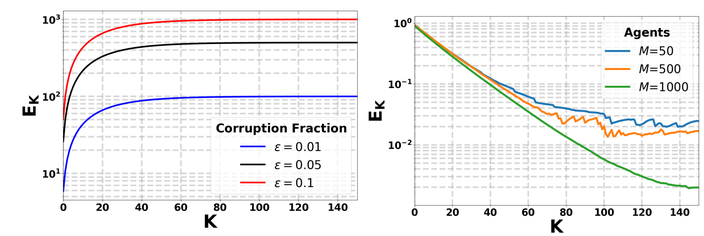
Main Results for Robust Fed-Q

$$e_K \leq \frac{e_0}{MT} + \tilde{\mathcal{O}}\left(\frac{1}{(1-\gamma)^{2.5}\sqrt{NT}}\right) + \mathcal{O}\left(\frac{\bar{R}\sqrt{\boldsymbol{\varepsilon}\log(NT)}}{(1-\gamma)^{2.5}\sqrt{T}}\right)$$

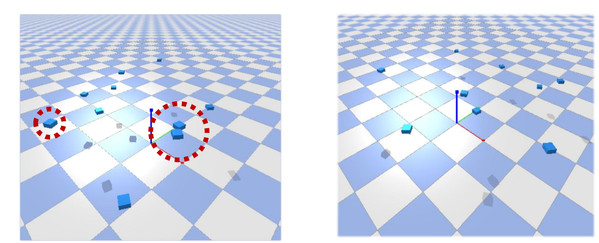
Main Takeaways:
- When $\varepsilon = 0$, our bound matches the **optimal** $\mathbf{1/\sqrt{NT}}$.
- Additive error ***does not*** depend on corruption magnitude.
- **Effect of corruption can be *drastically eliminated.***
- **Near-optimal statistical rates** can be achieved with just $\tilde{\mathcal{O}}(1)$ **rounds of communication.**

Simulations

Setup: Synthetic grid-world with 10 states, 5 actions, and $\gamma = 0.5$. We ran our experiments with $\alpha = 0.1$ for $T = 25x10^4$.



(Left) $E_k = \|Q_K - Q^*\|_\infty$ for $M = 1000$, where the central server naively averages the agents' updates. As we can see, only 10 adversarial agents in a pool of 1000 can completely skew the error metric. **(Right)** $E_k = \|Q_K - Q^*\|_\infty$ for Robust Fed-Q.



(Left) Simple area-coverage task with a team of robots exploring an environment. With naïve server averaging, agents collide and drift outside the designated area. **(Right)** Robust Fed-Q avoids these failures and successfully solves the task.

References:
[1] C. J. Watkins and P. Dayan, "Q-learning," Machine learning, vol. 8,pp. 279–292, 1992.
[2] Maity and Mitra, "Robust Federated Q-Learning with Almost No Communication", American Control Conference 2026.